The System Development Life Cycle

What is an information system (IS)?

Hardware, software, data, people, and procedures that work together to produce quality information

System—Set of components that interact to achieve common goal

Businesses use many types of systems

The System Development Life Cycle

What are the phases of the system development cycle?

Phase 1. Planning
- Review project requests
- Prioritize project requests
- Allocate resources
- Identify project development team

Phase 2. Analysis
- Conduct preliminary investigation
- Perform detailed analysis activities:
  - Study current system
  - Determine user requirements
  - Recommend solution

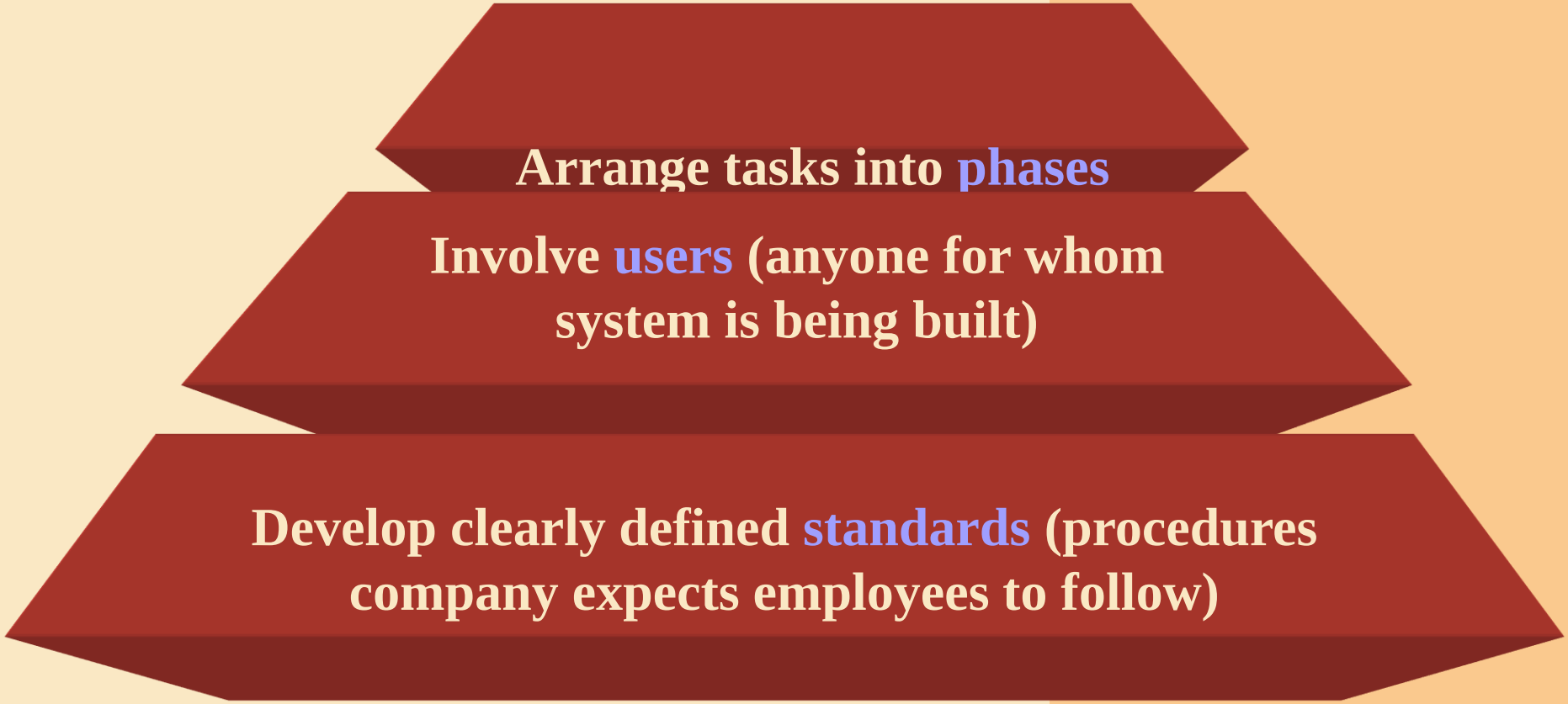
Phase 3. Design
- Acquire hardware and software, if necessary
- Develop details of system

Phase 4. Implementation
- Develop programs, if necessary
- Install and test new system
- Train users
- Convert to new system

Phase 5. Support
- Conduct post-implementation system review
- Identify errors and enhancements
- Monitor system performance

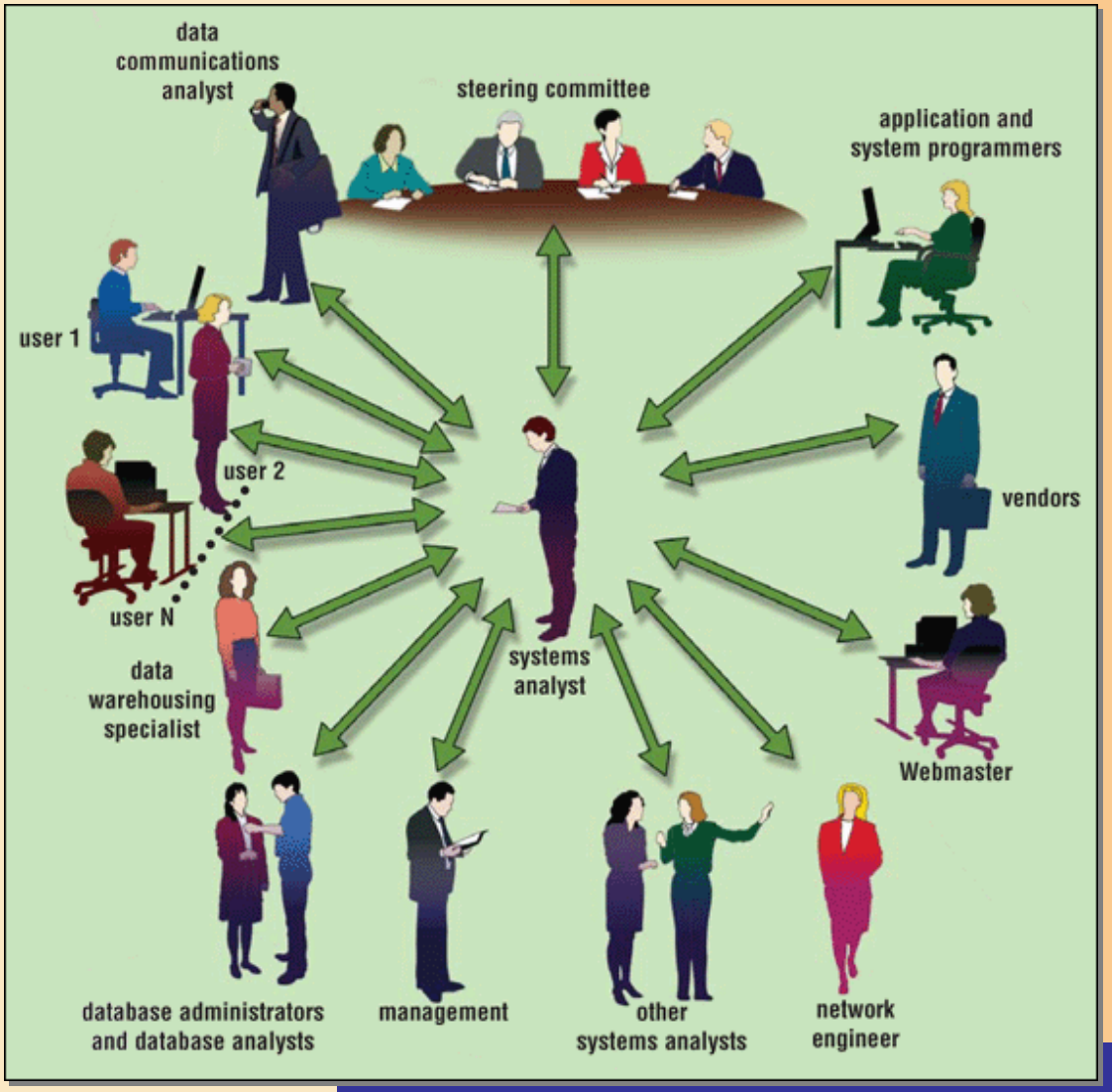
The System Development Life Cycle

What are guidelines for system development?



The System Development Life Cycle

Who participates in the system development life cycle?



The System Development Life Cycle

What is a systems analyst?

Responsible for designing and developing information system

Liaison between users and IT professionals

The System Development Life Cycle

What is the project team?

Formed to work on project from beginning to end

Consists of users, systems analyst, and other IT professionals

Project leader—one member of the team who manages and controls project budget and schedule

The System Development Life Cycle

What is feasibility?

Measure of how suitable system development will be to the company

Four feasibility tests:

- Operational feasibility
- Schedule feasibility
- Technical feasibility
- Economic feasibility (also called cost/benefit feasibility)

The System Development Life Cycle

What is documentation?

Collection and summarization of data and information

Includes reports, diagrams, programs, and other deliverables

The System Development Life Cycle

What are six data and information gathering techniques?

- **Review documentation**
- **Observe**
- **Questionnaire**
- **Interview**
- **Joint-application design (JAD) session**
- **Research**

The System Development Life Cycle

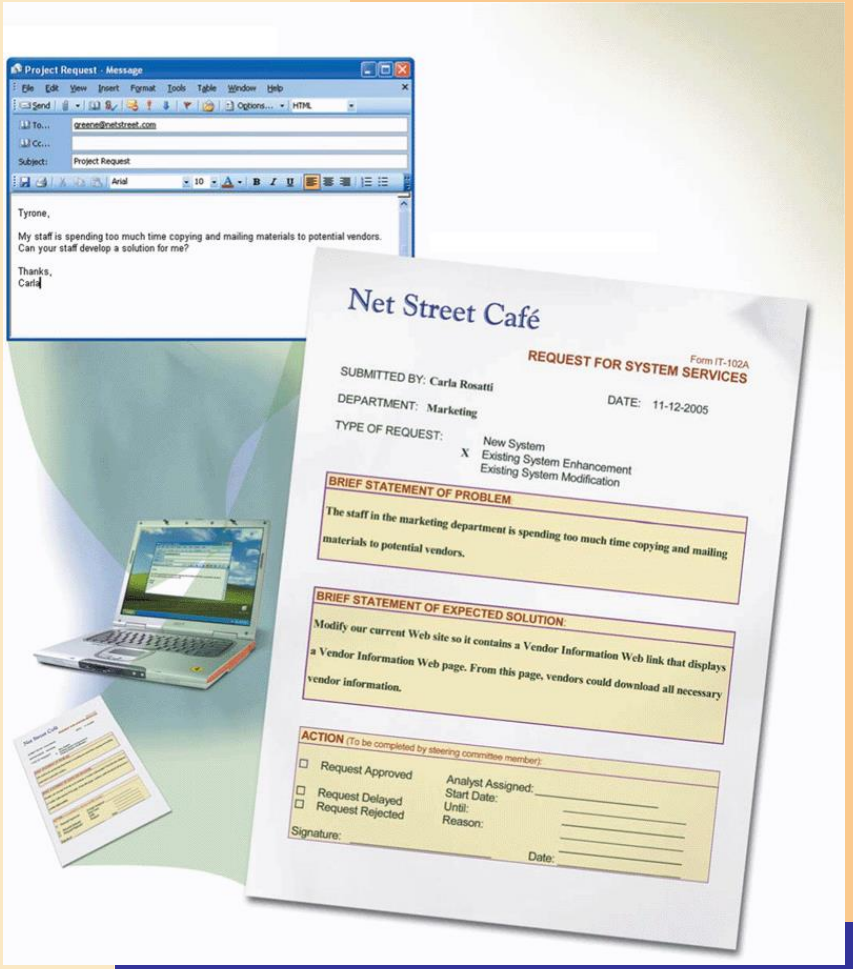
What are some reasons to create or modify an information system?

To correct problem in existing system	To improve existing system
Outside group may mandate change	Competition can lead to change

The System Development Life Cycle

What is a request for system services?

- **Formal request for new or modified information system**
  - Also called **project request**



The System Development Life Cycle

What is the planning phase?

Begins when steering committee receives project request

Steering committee—decision-making body for the company

Function of committee:

- **Review and approve project requests**
- **Prioritize project requests**
- **Allocate resources**
- **Form project development team for each approved project**

The System Development Life Cycle

What is the analysis phase?

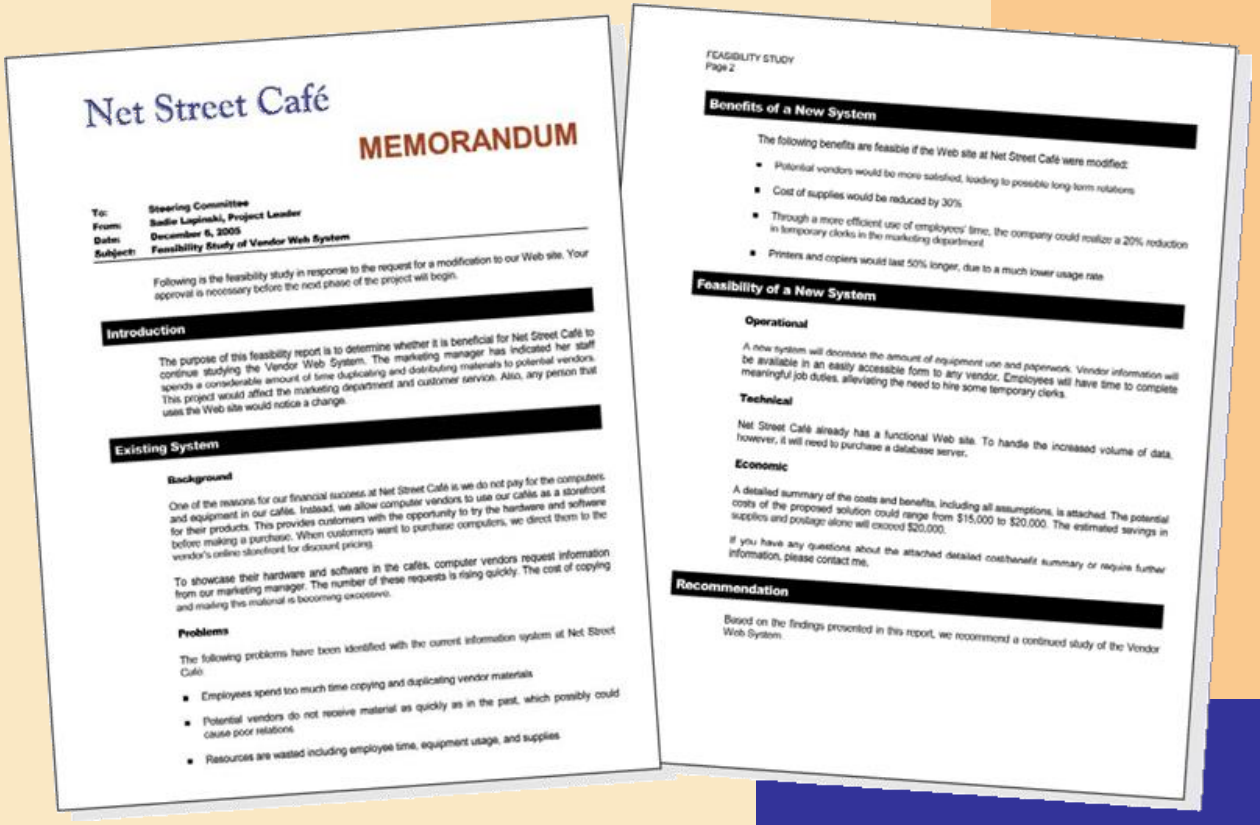
Conduct preliminary investigation, also called feasibility study

Perform detailed analysis

The System Development Life Cycle

What is the preliminary investigation?

- **Determine exact nature of problem or improvement and whether it is worth pursuing**
  - Findings are presented in feasibility report, also known as a feasibility study



Net Street Café

MEMORANDUM

To: Steering Committee
From: Sadie Lapinski, Project Leader
Date: December 6, 2005
Subject: Feasibility Study of Vendor Web System

Following is the feasibility study in response to the request for a modification to our Web site. Your approval is necessary before the next phase of the project will begin.

Introduction

The purpose of this feasibility report is to determine whether it is beneficial for Net Street Café to continue studying the Vendor Web System. The marketing manager has indicated her staff spends a considerable amount of time duplicating and distributing materials to potential vendors. This project would affect the marketing department and customer service. Also, any person that uses the Web site would notice a change.

Existing System

Background

One of the reasons for our financial success at Net Street Café is we do not pay for the computers and equipment in our cafés. Instead, we allow computer vendors to use our cafés as a storefront for their products. This provides customers with the opportunity to try the hardware and software before making a purchase. When customers want to purchase computers, we direct them to the vendor's online storefront for discount pricing.

To showcase their hardware and software in the cafés, computer vendors request information from our marketing manager. The number of these requests is rising quickly. The cost of copying and mailing this material is becoming excessive.

Problems

The following problems have been identified with the current information system at Net Street Café:

- Employees spend too much time copying and duplicating vendor materials
- Potential vendors do not receive material as quickly as in the past, which possibly could cause poor relations
- Resources are wasted including employee time, equipment usage, and supplies

FEASIBILITY STUDY
Page 2

Benefits of a New System

The following benefits are feasible if the Web site at Net Street Café were modified:

- Potential vendors would be more satisfied, leading to possible long-term relations
- Cost of supplies would be reduced by 30%
- Through a more efficient use of employees' time, the company could realize a 20% reduction in temporary clerks in the marketing department
- Printers and copiers would last 50% longer, due to a much lower usage rate

Feasibility of a New System

Operational

A new system will decrease the amount of equipment use and paperwork. Vendor information will be available in an easily accessible form to any vendor. Employees will have time to complete meaningful job duties, alleviating the need to hire some temporary clerks.

Technical

Net Street Café already has a functional Web site. To handle the increased volume of data, however, it will need to purchase a database server.

Economic

A detailed summary of the costs and benefits, including all assumptions, is attached. The potential costs of the proposed solution could range from $15,000 to $20,000. The estimated savings in supplies and postage alone will exceed $20,000.

If you have any questions about the attached detailed cost/benefit summary or require further information, please contact me.

Recommendation

Based on the findings presented in this report, we recommend a continued study of the Vendor Web System.

The System Development Life Cycle

What is detailed analysis?

1. Study how current system works

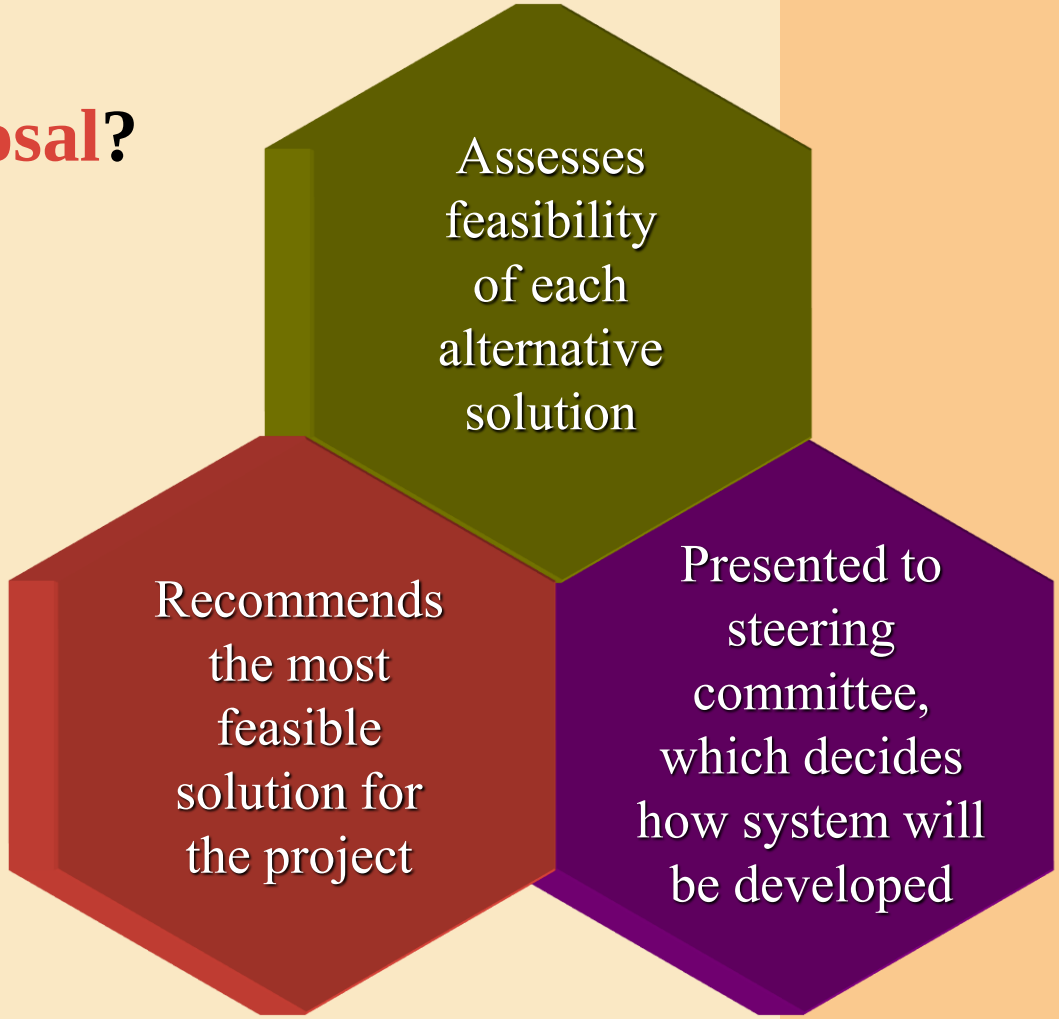
2. Determine user's wants, needs, and requirements

3. Recommend solution

Sometimes called logical design

The System Development Life Cycle

What is the system proposal?



The System Development Life Cycle

What are possible solutions?

Buy packaged software—prewritten software available for purchase

Write own custom software—software developed at user's request

Outsource—have outside source develop software

Horizontal market software—meets needs of many companies

Vertical market software—designed for particular industry

The System Development Life Cycle

What is the design phase?

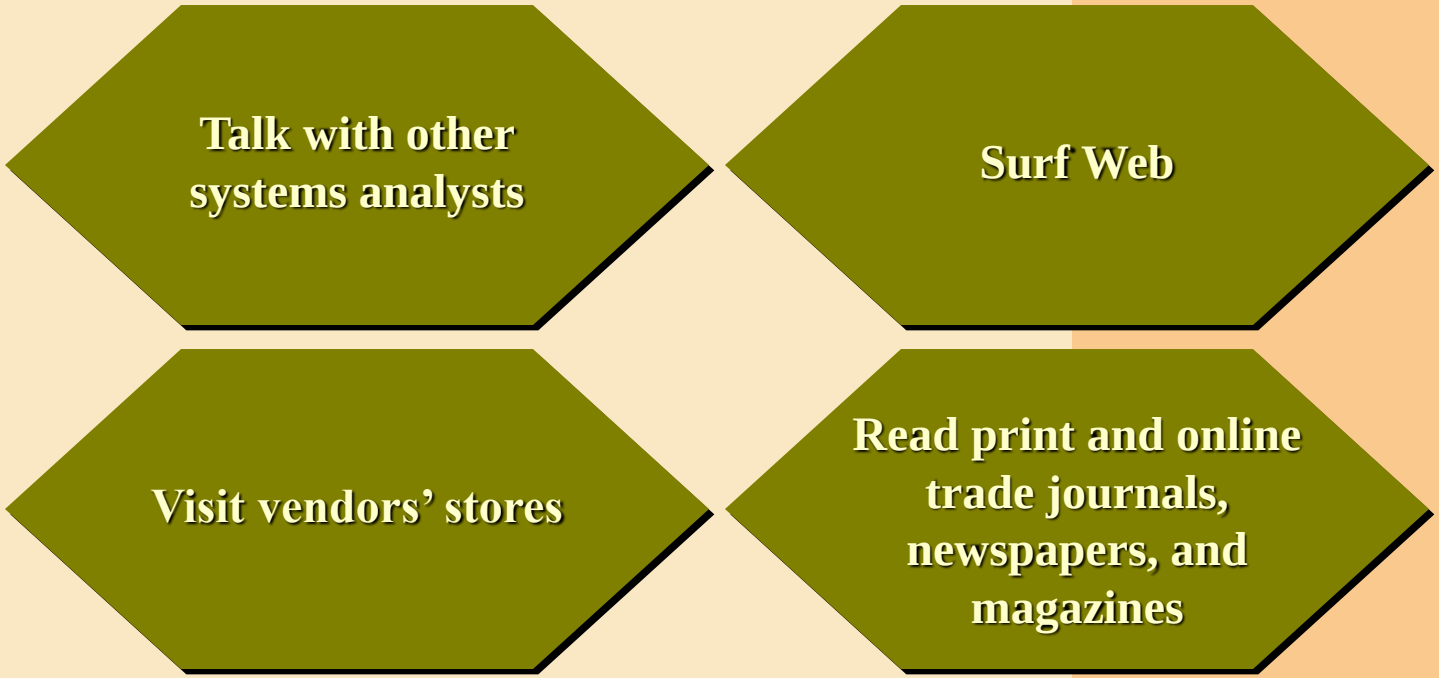
Acquire hardware and software

Develop all details of new or modified information system

The System Development Life Cycle

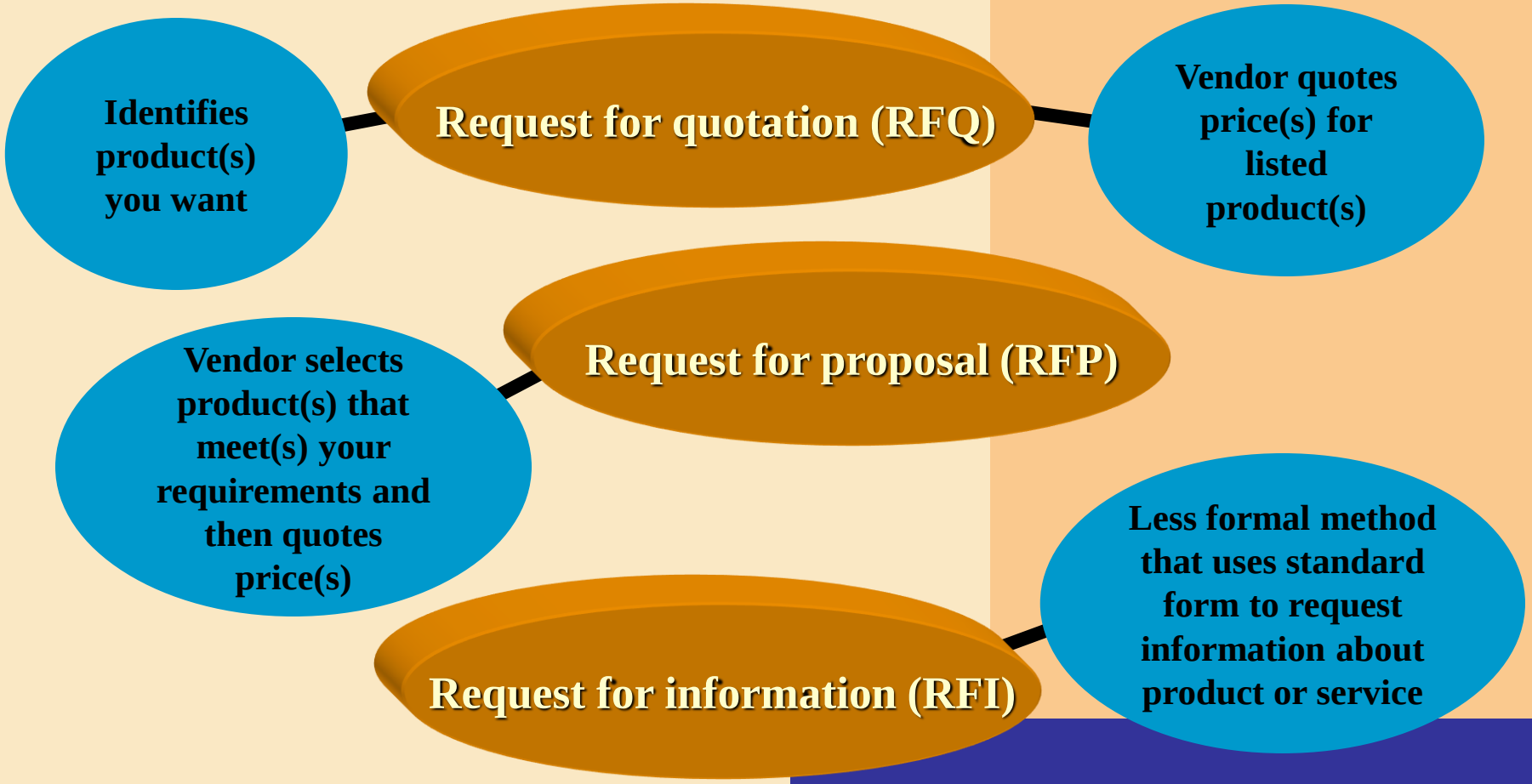
What is needed to acquire new hardware and software?

- Identify all hardware and software requirements of new or modified system



The System Development Life Cycle

What are three basic documents used to summarize technical specifications?



The System Development Life Cycle

How do systems analysts test software products?

- **References from vendor**
- **Talk to current users of product**
- **Product demonstrations**
- **Trial version of software**
- **Benchmark test measures performance**

The System Development Life Cycle

What is a detailed design?

Detailed design specifications for components in proposed solution

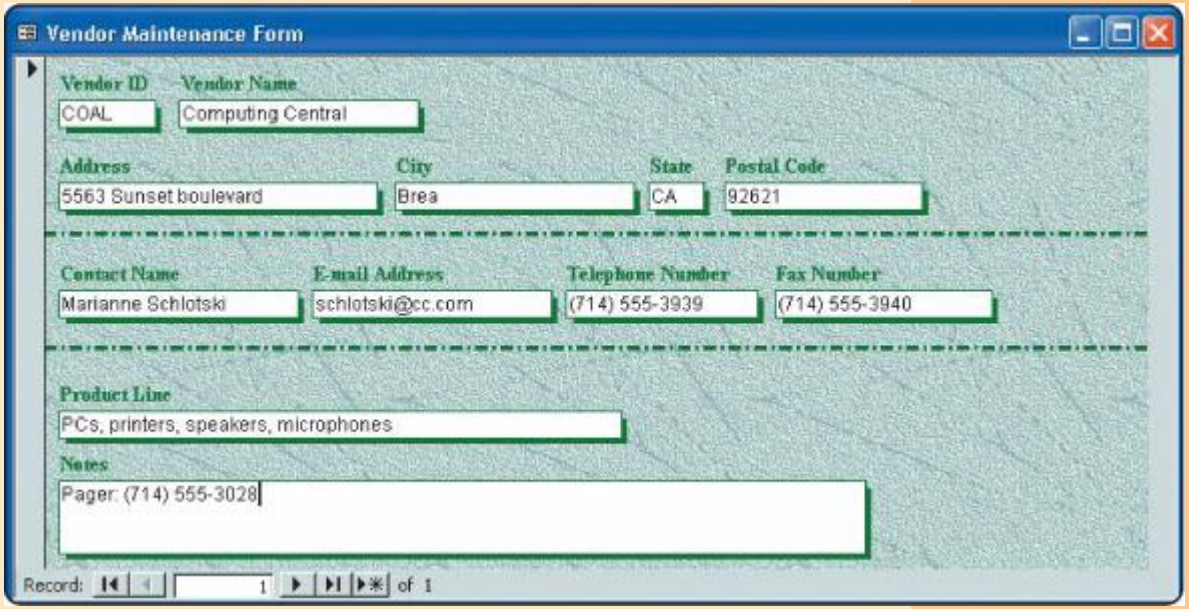
Includes several activities

- Database design
- Input and output design
- Program design

The System Development Life Cycle

What is a mockup?

- Sample of input or output that contains actual data



The System Development Life Cycle

What is a prototype?

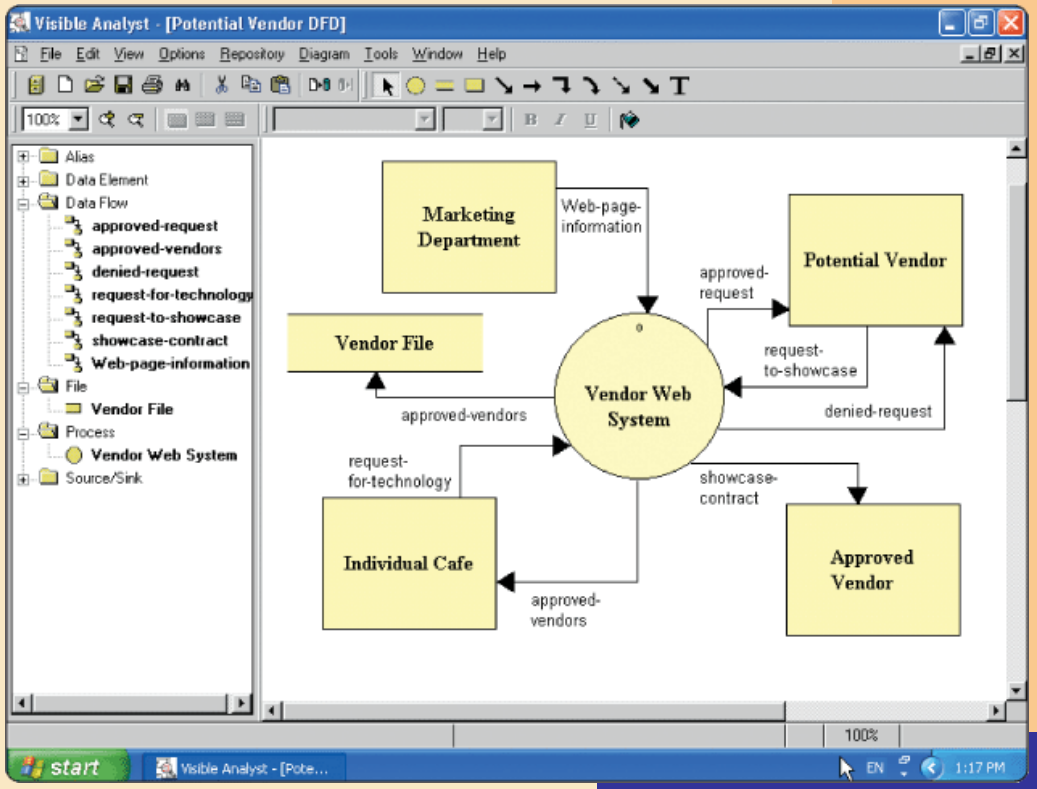
Working model of proposed system

Beginning a prototype too early may lead to problems

The System Development Life Cycle

What is computer-aided software engineering (CASE)?

- **Software tools designed to support activities of system development cycle**



The System Development Life Cycle

What is the implementation phase?

- **Purpose is to construct, or build, new or modified system and then deliver it to users**

The System Development Life Cycle

What are the three types of tests performed by system developers?

Unit Test

Verifies each individual program works by itself

Systems test

Verifies all programs in application work together

Integration Test

Verifies application works with other applications

The System Development Life Cycle

What is training?

- **Showing users exactly how they will use new hardware and software in system**

The System Development Life Cycle

What is the support phase?

- Provides ongoing assistance after system is implemented

Conduct post-implementation system review—meeting to find out if information system is performing according to expectations

Identify errors

Identify enhancements

Monitor system performance